

Quiz Justifications

Responder Link: <https://forms.gle/ZRaviPx EhH172LhS9>

Question 1. Correct Answer: To represent colours using combinations of Red, Green, and Blue

Why correct: RGB defines colours through additive mixing of red, green, and blue light, which is how digital displays generate all visible colours. Why others are wrong:

- Black-and-white images: RGB contains intensity info, but it is inherently designed for full colour.
- Reduce colours: confusion with compression methods, not colour representation.
- Internet speed: plausible due to “digital systems” association, but unrelated to networking.

Question 2. Correct Answer: 16,777,216

Why correct: 24-bit colour uses 8 bits per channel (256 levels each), so 256^3 gives 16,777,216 possible combinations. Why others are wrong:

- 256: only one channel (misunderstanding dimensionality).
- 65,536: confusion with 16-bit colour depth.
- 1,000,000: intuitive approximation, but colour spaces grow exponentially, not linearly.

Question 3. Correct Answer: Grey pixels have $S = 0$, so hue becomes undefined and defaults to 0°

Why correct: When saturation is zero, hue is mathematically undefined; implementations assign $H = 0^\circ$, which lies in the red range, causing false positives. Why others are wrong:

- Grey has hue: misconception—achromatic pixels have no hue.
- Tarmac reflection: irrelevant; issue is computational, not physical.
- OpenCV scaling: plausible assumption, but the bug arises from $S = 0$ handling, not scaling.

Question 4. Correct answer: Decrease S and V

Why correct: V controls brightness and S controls colour intensity; reducing both darkens and desaturates while keeping hue unchanged. Why others are wrong:

- Decrease S only: removes colour intensity but image stays bright.
- Decrease V and increase S: opposite effect (brighter image).
- Decrease S and increase V: conflicting adjustments, not achieving target effect.

Question 5. Correct Answer: Separates brightness from colour for efficient compression

Why correct: YCbCr isolates luminance (Y) from chroma (Cb/Cr), allowing compression to reduce colour data with minimal perceptual loss. Why others are wrong:

- Increases brightness: no enhancement function exists.
- More colours than RGB: same representational range.
- Removes compression need: it improves compression efficiency, not replaces it.

Question 6. Correct Answer: Ratio of luminance to chrominance samples

Why correct: Formats like 4:2:2 or 4:2:0 describe how colour information is reduced relative to brightness for bandwidth efficiency. Why others are wrong:

- Brightness reduction: luminance is typically fully preserved.
- RGB resolution: unrelated to encoding structure.
- Number of colours: sampling affects storage, not colour range.

Question 7. Correct Answer: LAB separates brightness from colour

Why correct: L^* encodes luminance while a^* and b^* encode colour, making segmentation robust under lighting changes. Why others are wrong:

- Fewer channels: LAB still uses 3 channels.
- Faster computation: conversion is more complex than RGB.
- Grayscale only: only L^* is grayscale, not full LAB space.

Question 8. Correct Answer: Opponent colour theory

Why correct: LAB is based on human vision, encoding colour as opposing pairs (red–green, blue–yellow), reflecting perceptual processing. Why others are wrong:

- Additive mixing: describes RGB, not LAB.
- Subtractive mixing: applies to CMYK printing.
- Binary representation: not a valid colour model concept.

Question 9. Correct Answer: $Y = 16$, $C_b = 128$, $C_r = 128$

Why correct: The +16 offset exists for a specific broadcast engineering reason. BT.601 uses limited range encoding: Y is offset to 16–235, while chroma is centred at 128 for broadcast standards. Why others are wrong:

- 0,0,0: ignores limited-range video encoding.
- 0,128,128: misses required luminance offset.
- 128,0,0: incorrect chroma centring and luminance placement.

Question 10. Correct Answer: HSV – Hue channel

Why correct: Hue isolates colour independent of lighting intensity, making it reliable for real-time object detection. Why others are wrong:

- RGB R channel: heavily affected by lighting changes.
- YCbCr C_r channel: still influenced by luminance.
- LAB a channel: accurate but computationally expensive for real-time use.